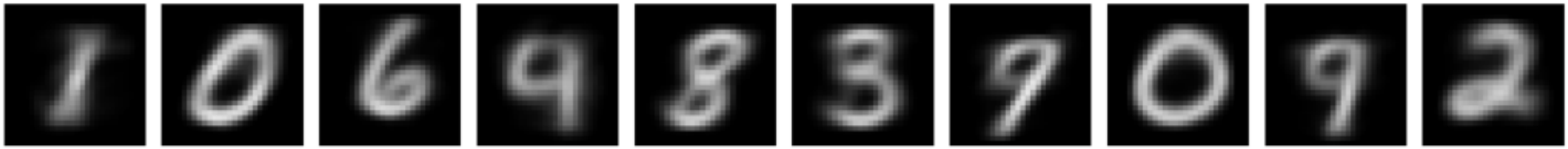
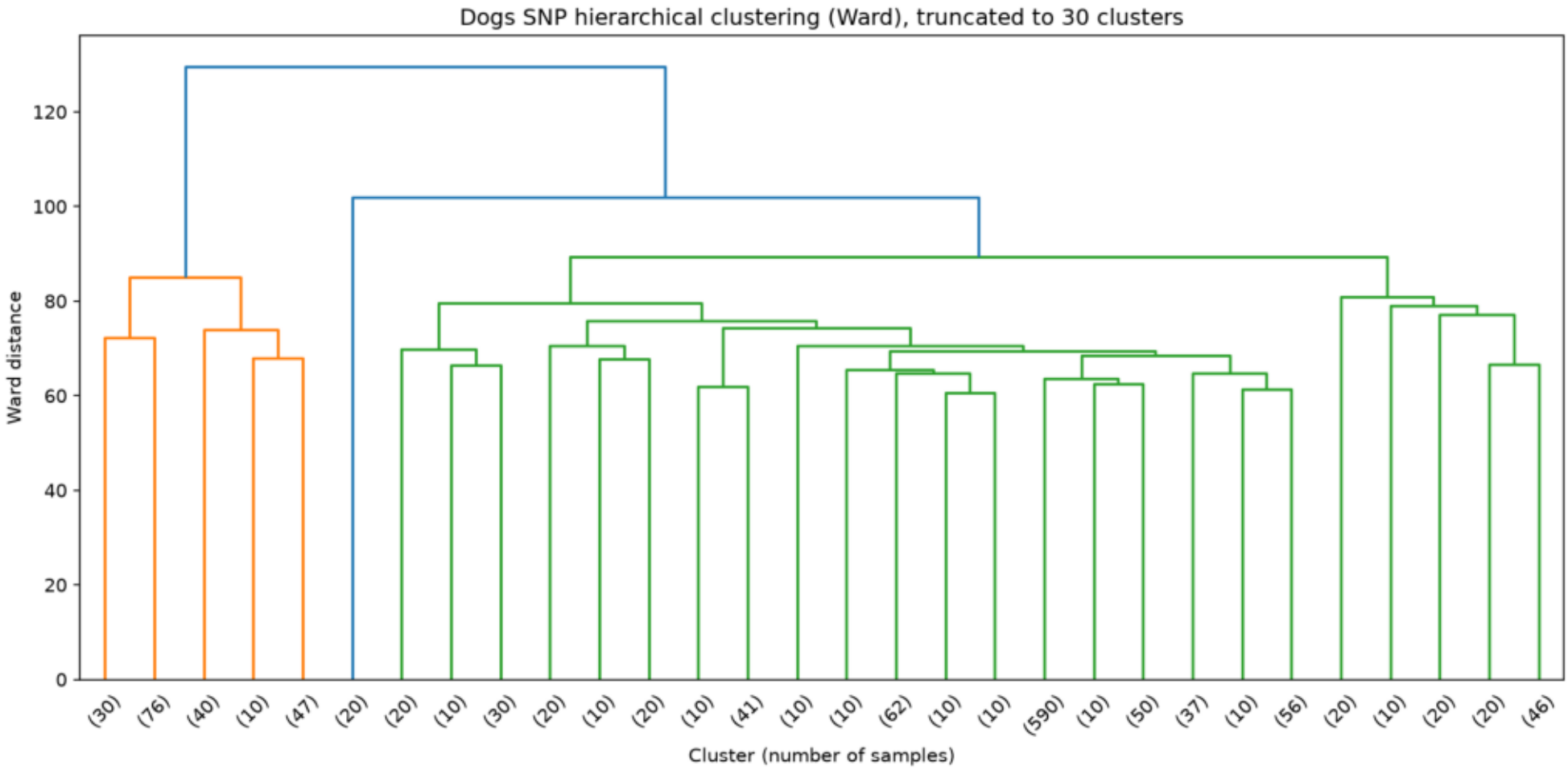
centroids_k10.png



centroids_k11.png



Dogs_dendrogram_truncated.png



Correctness evaluation

To make sure that my results are correct, I compared them with the results of scikit-learn, a standard Python library. For Part 1, I gave scikit-learn KMeans the same random start points (seeds 0-9) that my K-Means used, for k=10 and k=11. All 20 runs gave the same clustering ERROR and the same cluster labels (adjusted Rand index = 1.0). The ERROR changes from run to run only because the random start points change. For Part 2, scikit-learn AgglomerativeClustering (Ward linkage, 30 clusters) also gave k=30, ERROR=657, the same value as my SciPy script. This library calculates Ward linkage with different code than SciPy, thus the two results do not have a common source of error.